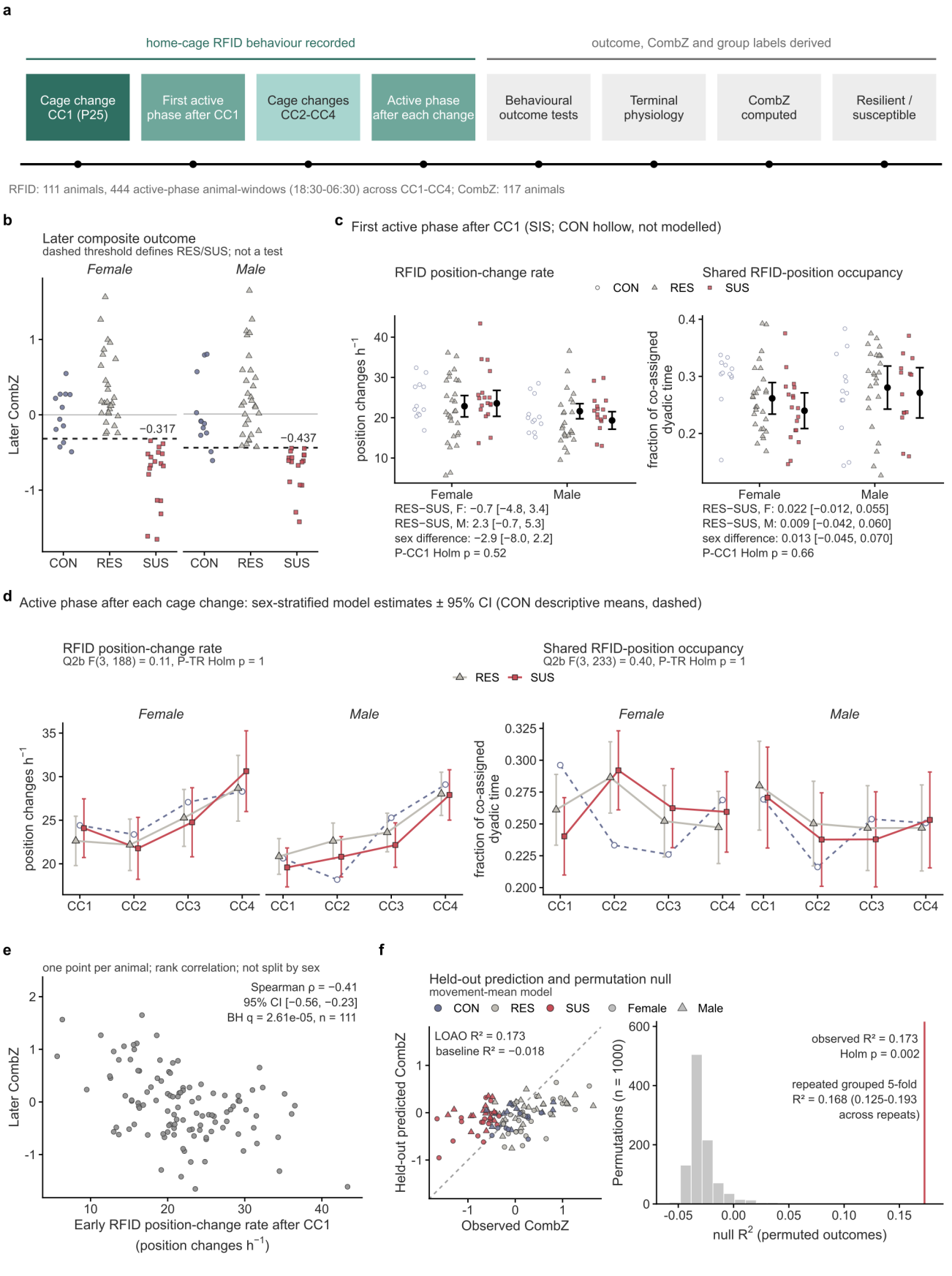

a
home-cage RFID behaviour recorded
outcome, CombZ and group labels derived
Cage change CC1 (P25)
First active phase after CC1
Cage changes CC2-CC4
Active phase after each change
Behavioural outcome tests
Terminal physiology
CombZ computed
Resilient / susceptible
RFID: 111 animals, 444 active-phase animal-windows (18:30-06:30) across CC1-CC4; CombZ: 117 animals
b
Later composite outcome
dashed threshold defines RES/SUS; not a test
Female
Male
Later CombZ
−0.317
−0.437
CON
RES
SUS
c
First active phase after CC1 (SIS; CON hollow, not modelled)
RFID position-change rate
Shared RFID-position occupancy
CON
RES
SUS
position changes h−1
fraction of co-assigned dyadic time
Female
Male
RES−SUS, F: −0.7 [−4.8, 3.4]
RES−SUS, M: 2.3 [−0.7, 5.3]
sex difference: −2.9 [−8.0, 2.2]
P-CC1 Holm p = 0.52
RES−SUS, F: 0.022 [−0.012, 0.055]
RES−SUS, M: 0.009 [−0.042, 0.060]
sex difference: 0.013 [−0.045, 0.070]
P-CC1 Holm p = 0.66
d
Active phase after each cage change: sex-stratified model estimates ± 95% CI (CON descriptive means, dashed)
RFID position-change rate
Q2b F(3, 188) = 0.11, P-TR Holm p = 1
Shared RFID-position occupancy
Q2b F(3, 233) = 0.40, P-TR Holm p = 1
RES
SUS
Female
Male
position changes h−1
fraction of co-assigned dyadic time
CC1
CC2
CC3
CC4
e
one point per animal; rank correlation; not split by sex
Spearman ρ = −0.41
95% CI [−0.56, −0.23]
BH q = 2.61e-05, n = 111
Later CombZ
Early RFID position-change rate after CC1
(position changes h−1)
f
Held-out prediction and permutation null
movement-mean model
CON
RES
SUS
Female
Male
LOAO R² = 0.173
baseline R² = −0.018
Held-out predicted CombZ
Observed CombZ
observed R² = 0.173
Holm p = 0.002
repeated grouped 5-fold
R² = 0.168 (0.125-0.193
across repeats)
Permutations (n = 1000)
null R² (permuted outcomes)